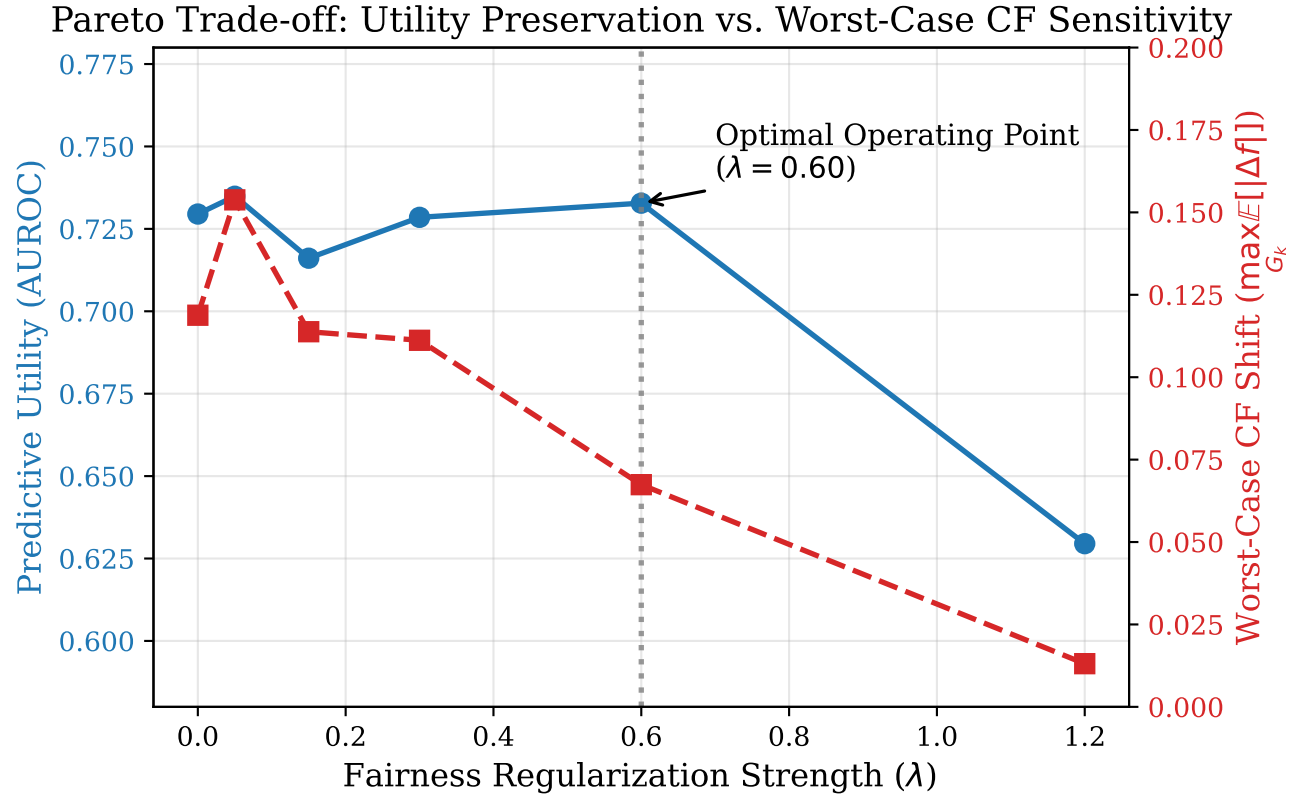
Pareto Trade-off: Utility Preservation vs. Worst-Case CF Sensitivity
Predictive Utility (AUROC)
Worst-Case CF Shift ($\max_{G_k} \mathbb{E}[|\Delta f|]$)
Fairness Regularization Strength (λ)
Optimal Operating Point
($\lambda = 0.60$)
0.775
0.750
0.725
0.700
0.675
0.650
0.625
0.600
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.0
0.2
0.4
0.6
0.8
1.0
1.2